

Technology #07: Self-Powered Sensing via Piezoelectric Energy Harvesting

Technology Domain: Piezoelectric Energy Harvesting + Implantable Sensing

Mechanism: Self-powered CSF pressure/flow sensor powered by harvested mechanical energy from CSF pulsations or catheter motion. R1 fix: original P-15 assumed unrealistic power density; R1 redesign uses piezoelectric transducer with rectifier + storage + duty-cycled sensing.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Harvested power at in-vivo strain levels is insufficient for any duty cycle (model shows <0.1 uW average)

Next Experiment: Piezo transducer (PVDF or PZT) on shaker table

Buyer Receives: Self-powered sensing concept + R1 correction (duty-cycled architecture); Piezo harvesting governing equations + power budget model; Bench test protocols (power, fatigue, biocompatibility); External precedent catalog (PVDF, PZT, lead-free piezo, harvesting ICs); Critical UNKNOWN disclosure (strain amplitude, chronic depoling, biocompatibility of piezo materials); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production piezo transducer design (material, geometry, integration with catheter); Power conditioning circuit (production PCB, miniaturized); Sensor selection + integration + self-referencing architecture; Manufacturing process for catheter-integrated piezo + circuit + sensor; Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2); Clinical validation (accuracy + longevity); Long-term reliability testing (fatigue, depoling, biocompatibility)

Transfer Posture: Sponsored Validation

Target Buyer: Implantable sensor company; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.